

Critical Cusps in Digital-Product Pressure

A Jordan collision, square-root phase scaling,
and explicit subcritical response

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Abstract

For digital masks in every integer base $b \geq 2$, we determine the leading phase singularity at the critical absolute-moment exponent $s = 1$:

$$P_{b,1}(c) = \sqrt{2(b-1)\log b} |c|^{1/2} + O_{b,\eta}(|c|^{1-\eta}), \quad 0 < \eta < \frac{1}{2}.$$

For the binary potential $\psi_c(x) = \log \cos^2 \pi(x - c)$ this is the previously excluded pressure order $q = 1/2$, since $s = 2q$. The square root comes from a genuine size-two Jordan block of the atomic transfer operator. We construct the generalized eigenfunction explicitly with the digamma function and prove an exponential remainder on every C^α , $0 < \alpha < 1$, by weighted quadrature rather than an assumed Perron theorem for a singular potential.

The same reduction gives a uniform two-parameter law when $s - 1 = u\sqrt{|c|}$ and an operator-norm finite-size crossover on the scale $n\sqrt{|c|}$. Separately, for every fixed $1/2 < s < 1$, we prove the exact subcritical cusp

$$P_{b,s}(c) = (1 - s) \log b + (b - 1)\pi s \tan(\pi s/2)|c| + o(|c|).$$

Its coefficient follows from a universal singular-translation integral. Together these results distinguish the critical square root, the subcritical linear cusp, and the repository's supercritical fractional response. Full proofs, numerical diagnostics, a proof-dependency audit, and a further-research agenda are included.

Contents

1	The research target and the scope of the answer	2
1.1	The published question and the repository gap	2
1.2	What is proved here	2
1.3	Provenance and novelty limits	2
2	Definitions and main theorems	3
2.1	Masks, transfer operators, and pressure	3
2.2	The critical square root	3
2.3	Detuning and finite size	4
2.4	The fixed subcritical range	4
3	Connecting cylinder pressure to spectral growth	5
4	The atomic operator has a genuine Jordan block	6
4.1	An eigenfunction and its generalized companion	6
4.2	Dual coordinates and the exact projection	7
4.3	Weighted quadrature and the rest of the spectrum	7
4.4	An exact critical normalization check	9
5	Perturbing the Jordan block	9
5.1	The norm of a phase translation	9
5.2	A second-order reduction lemma	9
5.3	The matrix entry that fixes the square root	10
5.4	Proof of the critical theorem	10
6	The two-parameter collision and its finite-time law	11
6.1	Why an analytic parameter basis matters	11
6.2	The effective generator	11
6.3	An operator, not only an eigenvalue, limit	12
7	Subcritical response from translated zeros	13
7.1	The dominant atomic eigenvalue is simple below one	13
7.2	A universal singular-translation integral	13
7.3	Converting the integral response to pressure	15
7.4	Why this proof stops at $s = 1/2$	15
8	The phase diagram and what the crossover explains	15
8.1	Three different responses at the same phase	15
8.2	Matching the singular coefficients	16
8.3	Which part is universal?	16
9	Reproducible numerical diagnostics	16
9.1	What the code does and does not verify	16
9.2	Critical scaled pressure	17
9.3	Subcritical linear response	17
9.4	Identity checks and finite-size behavior	18
9.5	Rebuilding the diagnostics	18
10	A further-research agenda	18
11	Conclusion	19
A	Proof-dependency and claim audit	19
B	Reproduction and package contents	20
	References	21

1 The research target and the scope of the answer

1.1 The published question and the repository gap

Gohlke, Kesseböhmer, and Schindler study generalized Thue–Morse measures and establish a spectral description of their pressure, including an analytic order-two case [1]. Gohlke, Lamprinakos, and Schmeling prove continuity in the phase for all nonnegative orders and explicitly ask about stronger regularity at other positive orders [2, Theorems 2.2–2.3 and the following discussion]. Their sine convention places the exceptional atomic phase at $1/2$; our equivalent cosine convention places it at zero.

The ProveIt repository contains two immediate predecessors. The integer-pressure report develops finite-dimensional methods at integer orders [4]. The fractional-pressure report proves a sharp atomic-phase normal form for all absolute-moment exponents $s > 1$ and explicitly leaves the critical and subcritical ranges untreated [5]. The present article addresses that excluded range. It does not repeat the integer-order optimization theorem or the supercritical regularity classification.

Our main target is not merely continuity at $s = 1$. We determine its exact leading singularity, its constant, its spectral mechanism, and its finite-size transition. A second argument determines the leading phase response for every fixed $1/2 < s < 1$. These are local answers at the atomic phase, not a claim to settle phase regularity everywhere.

1.2 What is proved here

The principal results are the critical square-root law (theorem 2.1), the detuned spectral and operator crossover (theorem 2.3), and the subcritical linear cusp (theorem 2.4). Their proofs include two structural results of independent use: an explicit Jordan decomposition with an exponential remainder (theorem 4.3), and a universal response formula for translated simple zeros (lemma 7.2).

The arguments rederive every repository-specific analytic ingredient needed for these results. No unverified repository theorem is used as an axiom. Standard facts about holomorphic functions, special functions, and Riesz projections are identified where used; the particular perturbation lemma needed here is proved as well.

Boundary	Precise status
Proved in this draft	Critical and detuned asymptotics in all bases; operator finite-size scaling; subcritical response for $1/2 < s < 1$.
Prior work	Atomic pressure at zero phase; positive-order continuity; order-two analyticity; the repository’s $s > 1$ phase analysis.
Not established	The phase asymptotic for $0 < s \leq 1/2$; a global phase regularity classification; full two-point endpoint Hölder regularity; worldwide priority; formal verification.

1.3 Provenance and novelty limits

The repository was inspected on 29 September 2026. The pinned head used for provenance is

06bcc37a82383ce99cebfec18f958487a5b3c6d4.

The fractional predecessor’s inspected source has Git blob identifier 83498d9ee7caa6f44b3a26b726cbb00c4b5a1a99. The contemporary papers [1, 2], the predecessor manuscripts, and targeted searches did not reveal a matching critical or subcritical formula. This is a bounded provenance check, not a guarantee that these results have never appeared elsewhere. The mathematical claim

is the proved statement below; a claim of an independently validated “breakthrough” would require additional scrutiny.

2 Definitions and main theorems

2.1 Masks, transfer operators, and pressure

Fix an integer $b \geq 2$ and let $T_b x = bx \pmod{1}$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Set

$$d_b(x) = \frac{1}{b} \sum_{r=0}^{b-1} e^{2\pi i r x}, \quad m_b(x) = |d_b(x)| = \left| \frac{\sin \pi b x}{b \sin \pi x} \right|, \quad (2.1)$$

with the removable values at integers. For $s > 0$ define

$$(\mathcal{L}_{s,c} f)(x) = \sum_{k=0}^{b-1} m_b \left(\frac{x+k}{b} - c \right)^s f \left(\frac{x+k}{b} \right). \quad (2.2)$$

The base is suppressed in $\mathcal{L}$ but not in the pressure. There is *no factor* $1/b$ in this transfer operator.

Let $\mathcal{D}_{b,n}$ be the b^n equal closed subintervals of $[0, 1]$. Writing $N = b^n$, put

$$F_{s,n,c}(x) = \prod_{j=0}^{n-1} m_b(b^j x - c)^s, \quad (2.3)$$

$$M_{s,n}(c) = \int_0^1 F_{s,n,c}(x) dx, \quad (2.4)$$

$$Z_{s,n}(c) = \sum_{I \in \mathcal{D}_{b,n}} \sup_{x \in I} F_{s,n,c}(x), \quad (2.5)$$

$$P_{b,s}(c) = \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_{s,n}(c). \quad (2.6)$$

We prove existence of this last limit in all the neighborhoods under consideration. A change of variables gives the exact normalization

$$\int_0^1 \mathcal{L}_{s,c}^n 1 dx = b^n M_{s,n}(c). \quad (2.7)$$

For the binary potential $\psi_c(x) = \log \cos^2 \pi(x - c)$,

$$P(q\psi_c) = P_{2,2q}(c). \quad (2.8)$$

In particular the critical exponent $s = 1$ means $q = 1/2$, not $q = 1$.

For $0 < \alpha < 1$, $C^\alpha(\mathbb{T})$ has norm $\|f\|_\alpha = \|f\|_\infty + [f]_\alpha$, with the circle distance in its Hölder seminorm. Spectral assertions refer to its complexification. We repeatedly use the constants

$$L = \log b, \quad a_b = 2 \log b, \quad d = b - 1, \quad \kappa_b = \sqrt{a_b d} = \sqrt{2(b-1) \log b}, \quad H(x) = \frac{|\sin \pi x|}{\pi}. \quad (2.9)$$

Here H is periodic, $H(x) \sim \text{dist}(x, \mathbb{Z})$, and $\int_0^1 H = 2/\pi^2$.

2.2 The critical square root

Theorem 2.1 (Critical phase law). *For every integer $b \geq 2$ and every $0 < \eta < 1/2$, the pressure is defined for all sufficiently small $|c|$ and*

$$P_{b,1}(c) = \kappa_b \sqrt{|c|} + O_{b,\eta}(|c|^{1-\eta}). \quad (2.10)$$

On C^η , the two eigenvalues near one are real and simple for small $c \neq 0$ and satisfy

$$\lambda_\pm(c) = 1 \pm \kappa_b \sqrt{|c|} + O_{b,\eta}(|c|^{1-\eta}). \quad (2.11)$$

The remaining spectrum is in a fixed disk of radius less than one, and $P_{b,1}(c) = \log \lambda_+(c)$.

Corollary 2.2 (Sharp pointwise regularity). *The even function $c \mapsto P_{b,1}(c)$ has exact pointwise Hölder exponent $1/2$ at zero. It is not Lipschitz there and has no finite first derivative there. Its right and left difference quotients tend to $+\infty$ and $-\infty$, respectively.*

The word *pointwise* is essential: this corollary asserts a bound against the value at zero and its optimality. It does not by itself prove a two-point $C^{0,1/2}$ bound throughout a neighborhood.

2.3 Detuning and finite size

Define, for $u \in \mathbb{R}$,

$$\omega_b(u) = \sqrt{\frac{u^2 L^2}{4} + \kappa_b^2}, \quad r_\pm(u) = -\frac{uL}{2} \pm \omega_b(u). \quad (2.12)$$

Let $\delta_0(f) = f(0)$; the rank-one operator $H \otimes \delta_0$ sends f to $Hf(0)$.

Theorem 2.3 (Uniform spectral and finite-size crossover). *Fix $0 < \alpha < 1/2$. Let $t > 0$, $s = 1 + ut$, and $c = \sigma t^2$, where $\sigma \in \{-1, 1\}$. Uniformly for u in any fixed compact subset of $\mathbb{R}$, as $t \downarrow 0$,*

$$\lambda_\pm(s, c) = 1 + tr_\pm(u) + o(t), \quad (2.13)$$

$$\frac{P_{b,1+ut}(\sigma t^2)}{t} \longrightarrow r_+(u). \quad (2.14)$$

More strongly, uniformly for (u, τ) in a compact subset of $\mathbb{R} \times [0, \infty)$, with $n = \lfloor \tau/t \rfloor$,

$$t \mathcal{L}_{1+ut, \sigma t^2}^n \longrightarrow \frac{a_b e^{-uL\tau/2} \sinh(\omega_b(u)\tau)}{\omega_b(u)} H \otimes \delta_0 \quad (2.15)$$

in operator norm on C^α . Consequently,

$$tb^n M_{1+ut, n}(\sigma t^2) \longrightarrow \frac{4 \log b}{\pi^2} e^{-uL\tau/2} \frac{\sinh(\omega_b(u)\tau)}{\omega_b(u)}. \quad (2.16)$$

This identifies $n \asymp |c|^{-1/2}$ as the finite-size transition scale. The uniformity is on compact u and τ sets; divergent detuning or divergent rescaled time is not included in the statement.

2.4 The fixed subcritical range

Theorem 2.4 (Explicit subcritical response). *For every fixed integer $b \geq 2$ and every fixed $1/2 < s < 1$,*

$$P_{b,s}(c) = (1-s) \log b + (b-1) \pi s \tan\left(\frac{\pi s}{2}\right) |c| + o(|c|). \quad (2.17)$$

Thus the two one-sided derivatives at the atomic phase are nonzero and opposite. In the usual binary pressure order, the range covered is $1/4 < q < 1/2$ and the coefficient is $2\pi q \tan(\pi q)$.

The restriction $s > 1/2$ is a rigorously identified boundary of the present perturbation estimate. It is *not* a claim that an additional thermodynamic transition occurs at $s = 1/2$.

3 Connecting cylinder pressure to spectral growth

A transfer eigenvalue need not automatically equal a cylinder-supremum pressure for a potential with zeros. We close that gap with a direct polynomial argument, valid even for $0 < s < 1$.

Lemma 3.1 (A uniform polynomial sampling inequality). *Let Q be a complex polynomial of degree at most $N - 1$, $N \geq 2$, and let $\mathcal{I}_N$ be the N equal cells of the circle. For every $s > 0$ there is a constant C_s , independent of N and Q , such that*

$$N \int_0^1 |Q(e^{2\pi i x})|^s dx \leq \sum_{I \in \mathcal{I}_N} \sup_I |Q(e^{2\pi i x})|^s \leq C_s N \int_0^1 |Q(e^{2\pi i x})|^s dx. \quad (3.1)$$

The constants can be bounded uniformly when s ranges over a compact subset of $(0, \infty)$.

Proof. The lower bound follows by integrating on each cell. For the upper bound, choose a maximizing point θ_I in every cell and set $R = 1 + 1/N$. Since $|Q|^s$ is subharmonic for every $s > 0$, the Poisson inequality on the disk of radius R gives, using angles in radians,

$$|Q(e^{i\theta_I})|^s \leq \int_0^{2\pi} P_{1/R}(\theta_I - \phi) |Q(Re^{i\phi})|^s \frac{d\phi}{2\pi}.$$

The Poisson kernel satisfies

$$P_{1/R}(\theta) \leq \frac{CN}{1 + N^2 \text{dist}(\theta, 2\pi\mathbb{Z})^2}.$$

There are only a bounded number of chosen points in each annular band of angular width $1/N$ about a fixed ϕ . This remains true if neighboring cells choose their common endpoint. Summing the convergent series $\sum_{j \geq 0} (1 + j^2)^{-1}$ yields $\sum_I P_{1/R}(\theta_I - \phi) \leq C'N$.

For $D = \deg Q$, consider the reversed polynomial $Q^*(z) = z^D \overline{Q(1/\bar{z})}$. On each circle, $|Q(Re^{i\phi})| = R^D |Q^*(R^{-1}e^{i\phi})|$. The integral means of the subharmonic function $|Q^*|^s$ are increasing with radius. Hence

$$\int |Q(Re^{i\phi})|^s \frac{d\phi}{2\pi} \leq R^{Ds} \int |Q(e^{i\phi})|^s \frac{d\phi}{2\pi} \leq e^s \int |Q(e^{i\phi})|^s \frac{d\phi}{2\pi}.$$

Combining the two estimates proves the upper bound with $C_s = C'e^s$. The zero polynomial is immediate. $\square$

In our setting,

$$Q_{n,c}(z) = b^{-n} \prod_{j=0}^{n-1} \left(\sum_{r=0}^{b-1} e^{-2\pi i r c} z^{rb^j} \right) \quad (3.2)$$

has degree exactly $b^n - 1$, and $F_{s,n,c}(x) = |Q_{n,c}(e^{2\pi i x})|^s$. Consequently,

$$b^n M_{s,n}(c) \leq Z_{s,n}(c) \leq C_s b^n M_{s,n}(c). \quad (3.3)$$

Whenever the logarithmic growth rate of the integral in equation (2.7) exists, so does the pressure, and their rates agree.

At zero phase the product telescopes:

$$F_{s,n,0}(x) = \left| \frac{\sin \pi b^n x}{b^n \sin \pi x} \right|^s. \quad (3.4)$$

Splitting into the $N = b^n$ cells and comparing distance from the two endpoints gives

$$NM_{s,n}(0) \asymp_s \begin{cases} N^{1-s}, & 0 < s < 1, \\ 1 + \log N, & s = 1, \\ 1, & s > 1. \end{cases} \quad (3.5)$$

For example, the cells at distance comparable to k/N contribute, after multiplication by N , quantities comparable to k^{-s} ; on a fixed positive fraction of each cell the numerator is bounded away from zero. Summing the resulting power series proves both bounds. It follows that

$$P_{b,s}(0) = \max\{(1-s) \log b, 0\}. \quad (3.6)$$

The binary version is already explicit in [1]; this derivation fixes the normalization and is not claimed as a new result.

4 The atomic operator has a genuine Jordan block

4.1 An eigenfunction and its generalized companion

Let $\psi = \Gamma'/\Gamma$ be the digamma function, and define on $(0, 1)$

$$G(x) = -H(x)(\psi(x) + \psi(1-x)), \quad G(0) = 1. \quad (4.1)$$

The endpoint expansion $\psi(x) = -x^{-1} - \gamma + O(x)$ shows that $G(x) = 1 + 2\gamma x + O(x^2)$ near zero; reflection gives the corresponding expansion near one. Thus G is a periodic Lipschitz function.

The following analytic family explains both the generalized eigenvector and the detuning parameter. Put

$$H_s = H^s, \quad \beta_s = b^{1-s}, \quad a_s = \frac{2(1-\beta_s)}{s-1}, \quad a_1 = 2 \log b, \quad (4.2)$$

and, initially for $s \neq 1$ close to one,

$$G_s(x) = H(x)^s \left(\zeta(s, x) + \zeta(s, 1-x) - \frac{2}{s-1} \right). \quad (4.3)$$

Here $\zeta(s, x)$ is the Hurwitz zeta function. Its pole at $s = 1$ has residue one, and its constant term is $-\psi(x)$, so $G_1 = G$. The standard special-function identities used here are recorded in [7]; their role is restricted to these explicit formulas.

Lemma 4.1 (Exact two-dimensional atomic family). *Fix $0 < \alpha < 1$. In a sufficiently small complex neighborhood of $s = 1$, H_s and G_s are analytic C^α -valued functions, with $H_s(0) = 0$ and $G_s(0) = 1$. Moreover,*

$$\mathcal{L}_{s,0} H_s = \beta_s H_s, \quad \mathcal{L}_{s,0} G_s = G_s + a_s H_s. \quad (4.4)$$

For real s the formulas use the usual nonnegative powers.

Proof. The coboundary identity

$$m_b(y) = \frac{H(T_b y)}{bH(y)} \quad (4.5)$$

away from the zeros immediately gives the first eigenfunction equation. For $\Re s > 1$, the periodization $S_s(x) = \sum_{j \in \mathbb{Z}} |x + j|^{-s}$ equals $\zeta(s, x) + \zeta(s, 1-x)$ on $(0, 1)$. Reindexing an absolutely convergent sum gives

$$\sum_{k=0}^{b-1} S_s \left(\frac{x+k}{b} \right) = b^s S_s(x).$$

Thus $\mathcal{L}_{s,0}(H_s S_s) = H_s S_s$. Subtracting $2H_s/(s-1)$ and using the first equation gives the second equation in equation (4.4).

For completeness, the analytic continuation is valid in the stated Banach space, not only pointwise. Near zero, write $S_s(x) = x^{-s} + A_s(x)$, with the pole $2/(s-1)$ removed from A_s . The regularized A_s is smooth in x on a one-sided neighborhood and analytic in s . The first part $(H(x)/x)^s$ is smooth up to the endpoint and equals one there. The remaining terms are x^s times smooth analytic factors. For s near one and $\Re s > \alpha$, these terms and all parameter derivatives are in C^α , uniformly on a smaller parameter disk. Factors $(\log x)^j$ from differentiation cause no loss because $\Re s - \alpha$ is bounded away from zero. The same argument works near one and for H_s . Analytic continuation proves the identities across $s = 1$ and supplies the endpoint values. $\square$

At $s = 1$, equation (4.4) reads

$$\mathcal{L}_{1,0}H = H, \quad (\mathcal{L}_{1,0} - \mathbf{I})G = a_b H. \quad (4.6)$$

The nonzero second equation is the source of the square root.

4.2 Dual coordinates and the exact projection

For s close enough to one, define

$$\ell_s(f) = \int_0^1 \frac{f(x) - f(0)G_s(x)}{H_s(x)} dx, \quad \ell_0(f) = f(0). \quad (4.7)$$

The notation ℓ_0 denotes evaluation, not the parameter value $s = 0$. We write $\ell = \ell_1$ at the critical exponent. These are bounded functionals on C^α : the numerator vanishes at zero and one, and the integrand is $O(d^{\alpha-\Re s})$, where $d = \text{dist}(x, \mathbb{Z})$. Choose $\Re s < 1 + \alpha$. The same estimates with powers of $\log d$ prove analyticity in s .

By direct calculation,

$$\ell_s(H_s) = 1, \quad \ell_s(G_s) = 0, \quad \ell_0(H_s) = 0, \quad \ell_0(G_s) = 1. \quad (4.8)$$

Also $\ell_0 \mathcal{L}_{s,0} = \ell_0$, since at $x = 0$ all inverse-branch weights except the branch at zero vanish. If $f(0) = 0$, a change of variables using equation (4.5) gives $\ell_s(\mathcal{L}_{s,0}f) = \beta_s \ell_s(f)$. Applying this to $f - f(0)G_s$ yields

$$\ell_s \mathcal{L}_{s,0} = \beta_s \ell_s + a_s \ell_0. \quad (4.9)$$

Therefore

$$\Pi_s f = H_s \ell_s(f) + G_s f(0) \quad (4.10)$$

is a commuting rank-two projection. On its range, in the ordered basis (H_s, G_s) , the atomic operator is exactly

$$A_s = \begin{pmatrix} \beta_s & a_s \\ 0 & 1 \end{pmatrix}. \quad (4.11)$$

4.3 Weighted quadrature and the rest of the spectrum

It remains to show that this finite-dimensional part captures the whole spectrum near one. A general spectral-gap theorem for a positive weight with zeros would need hypotheses not yet checked. The explicit product allows a direct proof instead.

Lemma 4.2 (Critical weighted quadrature). *If $0 < \alpha < 1$, $f \in C^\alpha$, and $f(0) = 0$, then*

$$\left\| \mathcal{L}_{1,0}^n f - H \int_0^1 \frac{f(y)}{H(y)} dy \right\|_\infty \leq C_{b,\alpha} (1+n) b^{-\alpha n} \|f\|_\alpha. \quad (4.12)$$

Proof. Let $N = b^n$. Telescoping the weights gives, for $0 < x < 1$,

$$\mathcal{L}_{1,0}^n f(x) = H(x) \frac{1}{N} \sum_{k=0}^{N-1} \frac{f((x+k)/N)}{H((x+k)/N)}. \quad (4.13)$$

Set $F = f/H$. Since $f(0) = 0$, one has $|F(y)| \leq C \|f\|_\alpha d(y)^{\alpha-1}$. The first and last cells have integral $O(N^{-\alpha})$. Their sampled terms, after multiplication by $H(x)$, have the same uniform bound: for instance the first is at most a constant times $N^{-\alpha} H(x) x^{\alpha-1}$, and this factor of x is bounded.

On an interior cell of length $1/N$, whose distance to an endpoint is comparable to $d \geq 1/N$, the difference of two values of F is bounded by

$$C \|f\|_\alpha (N^{-\alpha} d^{-1} + N^{-1} d^{\alpha-2}). \quad (4.14)$$

The first term uses Hölder continuity of f and the second uses $|H'| \leq 1$ together with $|f| \leq C \|f\|_\alpha d^\alpha$. Multiplying by the cell length and summing gives $O(\|f\|_\alpha N^{-\alpha}(1 + \log N))$. Since H is bounded, this proves the claimed uniform estimate. At $x = 0$ both sides of the main approximation are zero, so continuity covers the endpoint as well. $\square$

Theorem 4.3 (Exact critical Jordan decomposition). *On every C^α , $0 < \alpha < 1$, there are commuting finite-rank operators*

$$\Pi = H \otimes \ell + G \otimes \ell_0, \quad \mathcal{N} = a_b H \otimes \ell_0$$

and an operator R such that

$$\mathcal{L}_{1,0} = \Pi + \mathcal{N} + R, \quad \mathcal{N}^2 = 0, \quad \Pi \mathcal{N} = \mathcal{N} \Pi = \mathcal{N}, \quad \Pi R = R \Pi = \mathcal{N} R = R \mathcal{N} = 0. \quad (4.15)$$

For every r with $b^{-\alpha} < r < 1$ there is $C_r < \infty$ such that $\|R^n\|_{\alpha \rightarrow \alpha} \leq C_r r^n$ for $n \geq 1$. Consequently, for $n \geq 1$,

$$\mathcal{L}_{1,0}^n f = H(\ell(f) + n a_b f(0)) + G f(0) + R^n f. \quad (4.16)$$

The generalized eigenspace at one has dimension exactly two, and its Jordan block has size exactly two.

Proof. All algebraic assertions follow from equations (4.6) and (4.9). The invariant complement is

$$Y = \ker \Pi = \{f : f(0) = 0, \ell(f) = 0\}.$$

On Y , lemma 4.2 gives sup-norm decay $O((1+n)b^{-\alpha n})$. To upgrade to the full Hölder norm, first observe from equation (4.6) and the same quadrature estimate applied to $1 - G$ that

$$\|\mathcal{L}_{1,0}^m 1\|_\infty \leq C(1+m). \quad (4.17)$$

For each fixed m , pairing inverse branches in the circle gives

$$[\mathcal{L}_{1,0}^m f]_\alpha \leq b^{-\alpha m} \|\mathcal{L}_{1,0}^m 1\|_\infty [f]_\alpha + B_m \|f\|_\infty. \quad (4.18)$$

Indeed, the variation of f gains the inverse-branch contraction $b^{-\alpha m}$. The remaining term is bounded because the finitely many product weights are Lipschitz. For points across a circle cut, the branches are relabeled cyclically; for points separated by a fixed positive distance, the sup norm gives the same type of bound.

Given $b^{-\alpha} < r < 1$, choose m large enough that $C(1+m)b^{-\alpha m} < r^m$. Iterate equation (4.18) on Y , inserting equation (4.12) into its last term. The resulting scalar convolution of geometric sequences, with the harmless factor $1 + jm$, is bounded by a constant times r^{jm} . The finitely many remaining residues of n modulo m are absorbed into the constant. This proves the strong-norm bound for R^n .

The complement therefore has spectral radius at most $b^{-\alpha} < 1$. On the range of Π , the matrix is $\begin{pmatrix} 1 & a_b \\ 0 & 1 \end{pmatrix}$, with $a_b \neq 0$. No additional generalized eigenvector at one can occur in the complement. This proves the last assertion. $\square$

4.4 An exact critical normalization check

The functional on the constant function can also be evaluated:

$$\ell(1) = \int_0^1 \left(\frac{\pi}{\sin \pi x} + \psi(x) + \psi(1-x) \right) dx = 2 \log \frac{2}{\pi}. \quad (4.19)$$

To verify the cancellation, integrate from ε to $1 - \varepsilon$. The cosecant integral is $-2 \log \tan(\pi\varepsilon/2)$, while the digamma terms give $2(\log \Gamma(1 - \varepsilon) - \log \Gamma(\varepsilon))$. The endpoint expansions give equation (4.19). Integrating equation (4.16) at $f = 1$ yields

$$b^n M_{1,n}(0) = \frac{4 \log b}{\pi^2} n + C_* + O(r^n), \quad C_* = \int_0^1 G + \frac{4}{\pi^2} \log \frac{2}{\pi}. \quad (4.20)$$

This linear growth is the finite-size signature of the Jordan block. Its leading coefficient also agrees with the classical logarithmic integral of the telescoped Dirichlet-type kernel.

5 Perturbing the Jordan block

5.1 The norm of a phase translation

Lemma 5.1 (Phase perturbation bounds). *For fixed b and $0 < \alpha < 1$,*

$$\|\mathcal{L}_{1,c} - \mathcal{L}_{1,0}\|_{\alpha \rightarrow \alpha} = O(|c|^{1-\alpha}). \quad (5.1)$$

For fixed $0 < s < 1$ and $0 < \alpha < s$,

$$\|\mathcal{L}_{s,c} - \mathcal{L}_{s,0}\|_{\alpha \rightarrow \alpha} = O(|c|^{s-\alpha}). \quad (5.2)$$

If $s = 1 + ut$, $c = \pm t^2$, and u stays bounded, then uniformly

$$\|\mathcal{L}_{s,c} - \mathcal{L}_{s,0}\|_{\alpha \rightarrow \alpha} = O(t^{2-2\alpha}). \quad (5.3)$$

Proof. The mask is Lipschitz. Its translation difference has sup norm $O(|c|)$ and bounded Lipschitz seminorm. Interpolation gives a C^α bound $O(|c|^{1-\alpha})$. For $0 < s < 1$, the weight m_b^s is s -Hölder; the corresponding interpolation between the sup norm and the s -Hölder seminorm gives $O(|c|^{s-\alpha})$. Multiplication is bounded on C^α , as is composition with each inverse branch, so these weight estimates imply the operator estimates. For $s \geq 1$ near one, the Lipschitz constants are uniformly bounded. For $s < 1$ in the stated window, $t^{2(s-\alpha)} = t^{2-2\alpha} \exp(2ut \log t)$, and the final exponential factor is bounded uniformly for bounded u . $\square$

5.2 A second-order reduction lemma

We need the quadratic remainder in the following standard spectral reduction; merely a first-order error would not determine the critical coefficient. Riesz projection methods are treated systematically in [6]. We give the particular calculation to fix the coordinate convention and its error.

Lemma 5.2 (Compression in a Riesz-projection chart). *Let a bounded operator B_0 on a complex Banach space have a finite-rank commuting spectral projection P for an isolated spectral cluster. Write $E = \text{ran } P$, and let $A_0 = B_0|_E$ in a fixed basis. For $B = B_0 + \Delta$ with $\|\Delta\|$ small, let P_Δ be the Riesz projection defined by the same contour. The map $U = P_\Delta|_E : E \rightarrow \text{ran } P_\Delta$ is an isomorphism, and in this chart*

$$U^{-1}BU = A_0 + P\Delta|_E + O(\|\Delta\|^2). \quad (5.4)$$

The constants are uniform in a compact norm-continuous family with uniformly separated contours and uniformly bounded resolvents.

Proof. The resolvent identity and a Neumann series on the contour give $P_\Delta - P = O(\|\Delta\|)$. The identity $P_\Delta^2 = P_\Delta$, or its linearization, shows that

$$P(P_\Delta - P)P = O(\|\Delta\|^2).$$

Thus $PU = I_E + O(\|\Delta\|^2)$ and U is invertible onto the nearby spectral subspace. If $A = U^{-1}BU$, then

$$(PU)A = PBU = A_0(PU) + P\Delta U.$$

Here $PB_0 = B_0P$ and $P\Delta U = P\Delta|_E + O(\|\Delta\|^2)$. Multiplication by $(PU)^{-1}$ gives equation (5.4). Uniform bounds follow from the same contour estimates. $\square$

The usual separation of the complementary spectrum also follows directly from these estimates. A circle surrounding that spectrum and excluding the finite cluster remains a resolvent contour for small perturbations. Its resolvent integral bounds the powers of the complementary part uniformly by Cr^n for some $r < 1$ in our critical application.

5.3 The matrix entry that fixes the square root

Let $\Delta_{s,c} = \mathcal{L}_{s,c} - \mathcal{L}_{s,0}$. Evaluation at zero gives the following exact identity:

$$\ell_0(\Delta_{s,c}H_s) = \left| \frac{\sin \pi bc}{b\pi} \right|^s \sum_{k=1}^{b-1} \left(\frac{\sin(\pi k/b)}{|\sin \pi(k/b - c)|} \right)^s. \quad (5.5)$$

The branch $k = 0$ contributes zero because $H_s(0) = 0$. The denominators in the displayed finite sum are nonzero for small c . Reflection $k \mapsto b - k$ makes the sum even in c , so, uniformly for real s near one,

$$\ell_0(\Delta_{s,c}H_s) = (b-1)|c|^s(1 + O(c^2)). \quad (5.6)$$

In particular at $s = 1$ this is $d|c| + O(|c|^3)$. It is the *lower-left* entry in the ordered basis (H, G) . The nonzero upper-right entry a_b already exists at zero phase.

5.4 Proof of the critical theorem

Proof of theorem 2.1. Work on C^α with $\alpha = \eta \in (0, 1/2)$, and write $\varepsilon = |c|^{1-\alpha}$. By theorem 4.3, a small circle about one encloses exactly its two-dimensional generalized eigenspace. Apply lemma 5.2 with the projection equation (4.10). Its two coordinates are (ℓ, ℓ_0) , so equation (5.6) gives the central matrix

$$A(c) = \begin{pmatrix} 1 + O(\varepsilon) & a_b + O(\varepsilon) \\ d|c| + O(\varepsilon^2 + |c|^3) & 1 + O(\varepsilon) \end{pmatrix}. \quad (5.7)$$

The $O(\varepsilon^2)$ in the lower-left entry is the chart correction. All entries are real for real c : the contour projection can be chosen conjugation invariant.

Putting $z = \lambda - 1$, the characteristic equation is

$$z^2 - T(c)z - \kappa_b^2|c| + O(\varepsilon^2 + |c|^3) = 0, \quad T(c) = O(\varepsilon). \quad (5.8)$$

Because $\varepsilon = o(\sqrt{|c|})$ and $\varepsilon^2 = o(|c|)$, the discriminant is positive for small $c \neq 0$. The quadratic formula yields two distinct real roots. The error in their leading square roots is $O(\varepsilon^2/\sqrt{|c|} + |c|^{5/2})$, which is $O(\varepsilon)$ when $\alpha < 1/2$. This proves equation (2.11). The complementary spectrum and its powers remain uniformly inside a disk of radius less than one.

We must still show that the larger eigenvalue actually controls the integral of $\mathcal{L}^n 1$. Let $P_\pm(c)$ be the two simple eigenprojections. The two-dimensional formula

$$P_+(c)|_{\text{central}} = \frac{A(c) - \lambda_-(c)I}{\lambda_+(c) - \lambda_-(c)}$$

and its minus counterpart, together with the chart $U = I + O(\varepsilon)$, give

$$\int_0^1 P_{\pm}(c) 1 \, dx \sim \pm \frac{a_b}{\pi^2 \kappa_b \sqrt{|c|}}. \quad (5.9)$$

Indeed the coordinate of 1 along G tends to 1, the upper-right matrix entry tends to a_b , and $\int H = 2/\pi^2$. The coefficient for the larger eigenvalue is positive, not zero. For each fixed small $c \neq 0$, it follows that

$$\int \mathcal{L}_{1,c}^n 1 = \left(\int P_+(c) 1 \right) \lambda_+(c)^n + \left(\int P_-(c) 1 \right) \lambda_-(c)^n + O(r^n).$$

Thus its logarithmic growth rate is $\log \lambda_+(c)$. The sampling comparison equation (3.3) identifies this rate with the defined pressure. At $c = 0$ use equation (3.6). Finally $\log(1+z) = z + O(z^2)$ supplies equation (2.10). $\square$

Proof of corollary 2.2. Reflection $f(x) \mapsto f(-x)$ conjugates the operators at c and $-c$; the cylinder definition has the same symmetry. Thus the pressure is even. Since $P_{b,1}(0) = 0$ and $\kappa_b > 0$, equation (2.10) gives $P_{b,1}(c)/\sqrt{|c|} \rightarrow \kappa_b$. This proves all assertions, including failure of every pointwise Hölder exponent greater than $1/2$. $\square$

6 The two-parameter collision and its finite-time law

6.1 Why an analytic parameter basis matters

A direct comparison of $\mathcal{L}_{1+ut,c}$ with $\mathcal{L}_{1,0}$ has an $O(t)$ exponent perturbation. Squaring this error would be of the same size as the decisive t^2 lower-left entry. The exact s -dependent basis in lemma 4.1 avoids that loss: all exponent variation is treated exactly before the phase is perturbed.

Fix $0 < \alpha < 1/2$ and a small neighborhood of $s = 1$ with $\alpha < \Re s < 1 + \alpha$. The family $s \mapsto \mathcal{L}_{s,0}$ is analytic on C^α . To see this, differentiate the scalar weights m_b^s in s ; terms $m_b^s (\log m_b)^j$ remain in C^α on a smaller parameter disk because $\Re s > \alpha$. The projection Π_s in equation (4.10) is also analytic and commuting. Its complementary part has spectrum in a fixed disk $|z| < r < 1$ for s sufficiently close to one, by the gap proved at one and a resolvent Neumann series. Hence it is precisely the nearby central spectral projection.

6.2 The effective generator

Set $s = 1 + ut$, $c = \sigma t^2$, with bounded u . By lemma 5.1, the phase perturbation has norm $O(\varepsilon_t)$, where $\varepsilon_t = t^{2-2\alpha} = o(t)$. The compression lemma now applies relative to Π_s , uniformly in this window. In the ordered basis (H_s, G_s) the matrix is

$$A_{s,c} = A_s + \begin{pmatrix} \ell_s(\Delta_{s,c} H_s) & \ell_s(\Delta_{s,c} G_s) \\ \ell_0(\Delta_{s,c} H_s) & \ell_0(\Delta_{s,c} G_s) \end{pmatrix} + O(\varepsilon_t^2). \quad (6.1)$$

In addition to the norm estimate, the exact lower-left formula gives

$$\ell_0(\Delta_{s,c} H_s) = dt^2 \exp(2ut \log t) (1 + O(t^4)) = dt^2 + o(t^2) \quad (6.2)$$

uniformly for bounded u . Notice that $\varepsilon_t^2 = o(t^2)$ is exactly where $\alpha < 1/2$ is used.

Introduce the singular coordinate scaling $D_t = \text{diag}(t, 1)$. Since $\beta_{1+ut} = 1 - uLt + O(t^2)$ and $a_{1+ut} = a_b + O(t)$, equations (6.1) and (6.2) imply

$$D_t A_{s,c} D_t^{-1} = I + t J_b(u) + o(t), \quad J_b(u) = \begin{pmatrix} -uL & a_b \\ d & 0 \end{pmatrix}. \quad (6.3)$$

All errors are uniform on compact u sets. The characteristic polynomial of $J_b(u)$ is

$$z^2 + uLz - \kappa_b^2, \quad (6.4)$$

so its eigenvalues are $r_{\pm}(u)$ from equation (2.12). They are distinct for every real u ; one is positive and the other negative.

Proof of the spectral assertions in theorem 2.3. The two eigenvalues of equation (6.3) are $1 + tr_{\pm}(u) + o(t)$, uniformly on compact u sets. The cluster is real and simple because its rescaled limiting roots are distinct and real. The same projection-amplitude calculation as in equation (5.9) gives

$$t \int P_+(s, c) 1 \rightarrow \frac{a_b}{\pi^2 \omega_b(u)} > 0. \quad (6.5)$$

The rest of the spectrum stays uniformly below one in modulus, while $\lambda_+ > 1$ and $\lambda_- < 1$ for small t and bounded u . It follows from equation (3.3) that the pressure equals $\log \lambda_+$, and division by t proves equation (2.14). $\square$

6.3 An operator, not only an eigenvalue, limit

Proof of the finite-size assertions in theorem 2.3. Let $n = \lfloor \tau/t \rfloor$. From equation (6.3), elementary matrix product estimates give

$$(D_t A_{s,c} D_t^{-1})^n \rightarrow e^{\tau J_b(u)} \quad (6.6)$$

uniformly for (u, τ) in compact subsets of $\mathbb{R} \times [0, \infty)$. One proof compares with $(I + tJ_b(u))^n$ by a telescoping sum: the one-step error is $o(t)$, there are $O(1/t)$ steps, and both matrix products are uniformly bounded on a compact time interval. The usual exponential approximation then finishes the argument.

For an input f , its coordinates on the perturbed central space are $(\ell_s(f), f(0)) + O(\varepsilon_t \|f\|_{\alpha})$. After multiplication by D_t , these converge in operator norm to $(0, f(0))$. On output, the map from the rescaled coordinates back to t times the original function converges to $(v_1, v_2) \mapsto H v_1$. Indeed its columns are $H_s + O(\varepsilon_t)$ and $tG_s + O(t\varepsilon_t)$. Thus the limiting operator is the upper-right entry of $e^{\tau J_b(u)}$ multiplied by $H \otimes \delta_0$.

Since

$$\left(J_b(u) + \frac{uL}{2} I \right)^2 = \omega_b(u)^2 I,$$

that entry is

$$(e^{\tau J_b(u)})_{12} = a_b e^{-uL\tau/2} \frac{\sinh(\omega_b(u)\tau)}{\omega_b(u)}.$$

The complementary part contributes at most $Ct\tau^n$ in operator norm, which tends to zero uniformly even when $n = 0$ (then use its uniformly bounded projection). This proves equation (2.15). Applying the limit to 1, integrating, and using $\int H = 2/\pi^2$ proves equation (2.16). $\square$

At exact criticality $u = 0$, the moment law reduces to

$$tb^{\lfloor \tau/t \rfloor} M_{1, \lfloor \tau/t \rfloor}(\pm t^2) \rightarrow \frac{4 \log b}{\pi^2 \kappa_b} \sinh(\kappa_b \tau). \quad (6.7)$$

For small rescaled time its right side is $(4 \log b / \pi^2) \tau + O(\tau^3)$, in agreement with the atomic linear law equation (4.20). For larger fixed rescaled time it contains both exponentials coming from the split Jordan block. This explains why replacing the whole finite-size behavior by only its largest eigenvalue misses the crossover near the atomic phase.

7 Subcritical response from translated zeros

7.1 The dominant atomic eigenvalue is simple below one

For $0 < s < 1$, the eigenvalue $\beta_s = b^{1-s} > 1$ has right eigenfunction H_s and a particularly simple dual functional:

$$\nu_s(f) = \int_0^1 \frac{f(x)}{H_s(x)} dx, \quad \nu_s(H_s) = 1. \quad (7.1)$$

The integral is finite because $s < 1$, even for $f(0) \neq 0$. By a change of variables, $\nu_s \mathcal{L}_{s,0} = \beta_s \nu_s$.

Lemma 7.1 (Simple subcritical spectral gap). *Let $0 < s < 1$ and $0 < \alpha < s$. On C^α ,*

$$\beta_s^{-n} \mathcal{L}_{s,0}^n \longrightarrow H_s \otimes \nu_s \quad (7.2)$$

exponentially in operator norm. In particular β_s is a simple isolated eigenvalue and all other spectrum has strictly smaller modulus.

Proof. For $0 < x < 1$, telescoping gives

$$\beta_s^{-n} \mathcal{L}_{s,0}^n f(x) = H_s(x) \frac{1}{N} \sum_{k=0}^{N-1} \frac{f((x+k)/N)}{H_s((x+k)/N)}, \quad N = b^n. \quad (7.3)$$

The corresponding integral is $H_s \nu_s(f)$. Splitting into the endpoint cells and the interior cells, as in lemma 4.2, yields the uniform estimate

$$\|\beta_s^{-n} \mathcal{L}_{s,0}^n f - H_s \nu_s(f)\|_\infty \leq C(1+n)b^{-n \min\{\alpha, 1-s\}} \|f\|_\alpha. \quad (7.4)$$

Here is the difference from the critical calculation. The endpoint singularity is now d^{-s} , which is integrable without subtracting $f(0)$. Its endpoint contribution is $O(N^{s-1})$ after multiplication by $H_s(x)$, uniformly in x . The interior variation of the numerator contributes at most $CN^{-\alpha} \int_{1/N}^{1/2} y^{-s} dy$; the denominator variation contributes at most $CN^{-1} \int_{1/N}^{1/2} y^{-s-1} dy = O(N^{s-1})$. At $x = 0$, the normalized iterate is $\beta_s^{-n} f(0) = N^{s-1} f(0)$, so the same bound holds there. The extra factor $1+n$ in equation (7.4) is harmless and allows one bound for all estimates.

The normalized iterates of 1 are consequently uniformly bounded in sup norm. For each fixed m , the analogue of equation (4.18) has leading seminorm coefficient at most $Cb^{-\alpha m}$ and a finite sup-norm coefficient. The weights are s -Hölder, hence α -Hölder; this is sufficient for that estimate. On $\ker \nu_s$, combine the seminorm recurrence with equation (7.4). Choose $r < 1$ larger than both $b^{-\alpha}$ and $b^{-(1-s)}$, and then choose m large. The same geometric convolution argument as in theorem 4.3 proves exponential convergence in the full norm. The limit is the rank-one commuting projection in equation (7.2), proving the spectral assertions. $\square$

7.2 A universal singular-translation integral

The coefficient below is independent of the slopes of the zeros. This fact is what turns the digital-mask calculation into a simple multiple of the number of zeros.

Lemma 7.2 (Response of finitely many simple zeros). *Let m be a nonnegative periodic function, positive and C^2 away from N_z zeros. Near each zero z_j , assume that $m(z_j + x) = |x|a_j(x)$ with a_j positive and C^2 . For every $0 < s < 1$,*

$$\int_0^1 \left[\left(\frac{m(x-c)}{m(x)} \right)^s - 1 \right] dx = N_z J_s |c| + o(|c|), \quad (7.5)$$

where

$$J_s = \int_0^\infty u^{-s} ((u+1)^s + |u-1|^s - 2u^s) du \quad (7.6)$$

$$= \pi s \tan\left(\frac{\pi s}{2}\right) > 0. \quad (7.7)$$

Values assigned at the finitely many zero denominators do not affect the integral, whose singularities are integrable.

Proof. Choose small disjoint symmetric neighborhoods of the zeros, with their closures still inside the stated local charts. On the complement, ordinary Taylor expansion gives the first-order integral $-sc \int (\log m)' + o(|c|)$.

In a neighborhood $[-r, r]$ of a zero, write

$$\left(\frac{m(x-c)}{m(x)}\right)^s = \frac{|x-c|^s}{|x|^s} \left(\frac{a(x-c)}{a(x)}\right)^s.$$

Uniformly on this fixed neighborhood,

$$\left(\frac{a(x-c)}{a(x)}\right)^s = 1 - sc \frac{a'(x)}{a(x)} + O(c^2).$$

The ratios $|x-c|^s/|x|^s$ have bounded integrals and converge to one in $L^1([-r, r])$. Hence the local contribution is

$$\int_{-r}^r \left(\frac{|x-c|^s}{|x|^s} - 1\right) dx - sc \int_{-r}^r \frac{a'(x)}{a(x)} dx + o(|c|). \quad (7.8)$$

In the first integral substitute $x = |c|u$ and pair positive and negative u . The result divided by $|c|$ tends to equation (7.6). The paired integrand is $O(u^{-s})$ near zero and $O(u^{-2})$ at infinity, so its integral converges throughout the stated range. The sign of c does not change this paired model integral.

All the regular first-order contributions cancel. Indeed, in a local chart $(\log m)' = 1/x + a'/a$, and the principal value of $1/x$ on a symmetric interval is zero. Periodicity gives

$$\text{PV} \int_0^1 (\log m)' dx = 0.$$

More explicitly, after deleting symmetric intervals of radius ε around the zeros, the sum of endpoint logarithms has terms $\log(\varepsilon a_j(-\varepsilon)) - \log(\varepsilon a_j(\varepsilon))$, each tending to zero. Combining equation (7.8) with the complement thus proves equation (7.5) with coefficient $N_z J_s$.

To evaluate the coefficient, substitute $u = 1/v$ in equation (7.6), and integrate by parts. Both endpoint terms vanish, yielding

$$\frac{J_s}{s} = \int_0^\infty \frac{(1+v)^{s-1} + \text{sgn}(v-1)|v-1|^{s-1}}{v} dv. \quad (7.9)$$

The divergent pieces at zero must remain paired. In the first positive piece use $t = v/(1+v)$, and pair it with the negative piece on $(0, 1)$. Their sum is

$$\int_0^1 \frac{(1-t)^{-s} - (1-t)^{s-1}}{t} dt = \psi(s) - \psi(1-s) = -\pi \cot(\pi s).$$

The remaining positive piece is

$$\int_1^\infty \frac{(v-1)^{s-1}}{v} dv = B(s, 1-s) = \pi \csc(\pi s).$$

The last identities are the beta and digamma reflection identities [7, §5.5]. Therefore $J_s = \pi s(\csc \pi s - \cot \pi s) = \pi s \tan(\pi s/2)$, proving equation (7.7). $\square$

7.3 Converting the integral response to pressure

Proof of theorem 2.4. Fix $1/2 < s < 1$ and choose

$$0 < \alpha < s - \frac{1}{2}. \quad (7.10)$$

The simple eigenvalue in lemma 7.1 can be perturbed on C^α . The rank-one case of lemma 5.2, together with equation (5.2), gives

$$\lambda_s(c) = \beta_s + \nu_s((\mathcal{L}_{s,c} - \mathcal{L}_{s,0})H_s) + O(|c|^{2(s-\alpha)}). \quad (7.11)$$

The exponent in the error is greater than one by equation (7.10).

Using the coboundary equation (4.5) and changing variables over the inverse branches, one obtains the exact response identity

$$\nu_s((\mathcal{L}_{s,c} - \mathcal{L}_{s,0})H_s) = \beta_s \int_0^1 \left[\left(\frac{m_b(y-c)}{m_b(y)} \right)^s - 1 \right] dy. \quad (7.12)$$

All singularities in this integral are integrable because $s < 1$. The mask m_b has exactly $b - 1$ simple zeros on the circle, at $1/b, \dots, (b-1)/b$. Its absolute-value local form has positive smooth factors, so lemma 7.2 applies. Consequently

$$\lambda_s(c) = \beta_s(1 + (b-1)J_s|c| + o(|c|)). \quad (7.13)$$

The spectral projection of the constant function has positive integral at $c = 0$, namely $\nu_s(1) \int H_s > 0$, and retains this property for small c . All other spectrum stays strictly smaller in modulus. Therefore the logarithmic growth rate of $\int \mathcal{L}_{s,c}^n 1$ is $\log \lambda_s(c)$. The pressure comparison identifies this with $P_{b,s}(c)$. Taking logarithms in equation (7.13), using $\log \beta_s = (1-s) \log b$ and equation (7.7), proves the theorem. $\square$

7.4 Why this proof stops at $s = 1/2$

The scalar translation integral equation (7.5) remains valid for every $0 < s < 1$. The missing step for $s \leq 1/2$ is not the integral's evaluation, nor the simple atomic spectral gap. It is the comparison of the quadratic operator remainder with the proposed linear term. On C^α the available bound is $O(|c|^{2(s-\alpha)})$, and no positive α makes this $o(|c|)$ when $s \leq 1/2$.

This isolates a concrete research problem: replace the generic norm-square remainder by a sharper estimate that retains the localization of the translated zeros. The present paper does not infer a linear response below the threshold merely from its formally correct first-order coefficient. In particular, the formula at $q = 1/4$ is not claimed.

8 The phase diagram and what the crossover explains

8.1 Three different responses at the same phase

The results can be compared without changing conventions:

Exponent	Leading local information	Source/status
$1/2 < s < 1$	$P(c) - P(0) \sim (b-1)\pi s \tan(\pi s/2) c $	Theorem 2.4.
$s = 1$	$P(c) \sim \sqrt{2(b-1) \log b} \sqrt{ c }$	Theorem 2.1.
$s > 1$	$P(c) = s \log D_b(c) + 2(b^s - 1)\zeta(s) c ^s + o(c ^s)$	Repository predecessor [5].

Here $D_b(c) = \sin(\pi bc)/(b \sin \pi c)$ is positive near zero. In the third row the analytic term must not be dropped: when $s > 2$ it can dominate the displayed fractional term, and at even integer s coefficient cancellations are important. That row records the predecessor's statement, not a theorem newly reproved in this article.

The atomic spectral picture distinguishes the first two rows. Below one, $\beta_s > 1$ is a simple dominant eigenvalue with integrable dual density H^{-s} . At one, that density reaches its integrability threshold and a generalized eigenvector is required instead. The two central eigenvalues β_s and 1 coalesce, while the upper-right entry has the nonzero limit $2 \log b$. Opening a phase creates a lower-left entry of size $(b-1)|c|$. Their product produces the quadratic equation $z^2 \sim 2(b-1) \log b |c|$.

8.2 Matching the singular coefficients

As $s \uparrow 1$,

$$(b-1)\pi s \tan(\pi s/2) \sim \frac{2(b-1)}{1-s}. \quad (8.1)$$

The predecessor's coefficient has the corresponding pole from above:

$$2(b^s - 1)\zeta(s) \sim \frac{2(b-1)}{s-1} \quad (s \downarrow 1). \quad (8.2)$$

These divergent fixed-exponent coefficients signal that those asymptotics cannot be uniform across the critical point. The crossover function has the consistent tails

$$r_+(u) \sim \frac{2(b-1)}{u} \quad (u \rightarrow +\infty), \quad (8.3)$$

$$r_+(u) = -u \log b + \frac{2(b-1)}{-u} + O(|u|^{-3}) \quad (u \rightarrow -\infty). \quad (8.4)$$

This is an algebraic consistency check between the formulas. It is not an interchange-of-limits theorem: theorem 2.3 was proved only for bounded u . Establishing a uniform overlap region with $u \rightarrow \infty$ would require additional remainder estimates.

8.3 Which part is universal?

Two constants have transparent origins. The factor $b-1$ counts the simple zeros that feed the atomic fixed-point branch when the mask is translated. The factor $2 \log b$ is the nilpotent coupling in the critical Jordan chain. The subcritical constant J_s is a local translation invariant independent of the slopes of those zeros.

This does not establish universality for arbitrary masks: a different mask need not possess the coboundary equation (4.5), the same critical exponent, or a two-dimensional Jordan block. What the proof provides is a testable mechanism. If an analogous isolated Jordan chain, a sufficiently small perturbation remainder, and a nonzero lower-left response can be proved for another model, the same finite-dimensional calculation determines its leading splitting.

9 Reproducible numerical diagnostics

9.1 What the code does and does not verify

The accompanying code/verify.py has four roles. It checks explicit special-function identities at high precision; verifies the exact characteristic polynomial of the 2×2 generator with symbolic algebra; computes transfer-matrix collocation diagnostics; and tests the finite-size crossover by direct matrix iteration. Its actual output is included as data/verification.json and data/run.log.

The collocation matrix uses a uniform periodic grid and piecewise-linear interpolation of inverse-branch evaluations. It is sparse and nonnegative. Leading eigenvalues are computed by ARPACK with a fixed starting vector, with residuals recorded. The main critical grid has 65 536 points. Subcritical rows with $|c| = 10^{-5}$ use 262 144 points because dividing a pressure difference by $|c|$ amplifies grid error substantially.

These are diagnostics, not certified enclosures of the spectrum of an infinite-dimensional operator. A small matrix residual certifies neither interpolation accuracy nor a limiting theorem. None of the analytic proofs depends on the output of this program.

9.2 Critical scaled pressure

The values in the third column below approximate $P_{b,1}(c)/\sqrt{|c|}$. The last column is the theorem's predicted limit.

b	$ c $	$P_{b,1}(c)/\sqrt{ c }$	Predicted limit
2	10^{-2}	0.90534238	1.17741002
2	10^{-3}	1.02347259	1.17741002
2	10^{-4}	1.10259590	1.17741002
2	10^{-5}	1.14556562	1.17741002
3	10^{-2}	1.58772057	2.09629415
3	10^{-3}	1.80542336	2.09629415
3	10^{-4}	1.95254663	2.09629415
3	10^{-5}	2.03412370	2.09629415
5	10^{-2}	2.66625974	3.58824516
5	10^{-3}	3.05174974	3.58824516
5	10^{-4}	3.31691329	3.58824516
5	10^{-5}	3.46900368	3.58824516

The approach to the limit is visibly slow, especially at larger base. This is compatible with the proved error but does not measure its implicit constant. The data are reported as computed, rather than extrapolated values presented as certified constants.

9.3 Subcritical linear response

For the binary base, the corresponding diagnostic is $(P_{2,s}(c) - (1 - s) \log 2)/|c|$:

s	$ c $	$(P_{2,s}(c) - (1 - s) \log 2)/ c $	Predicted limit
0.60	10^{-3}	2.4819902	2.5944188
0.60	10^{-4}	2.5778127	2.5944188
0.60	10^{-5}	2.5952322	2.5944188
0.75	10^{-3}	5.2217464	5.6883567
0.75	10^{-4}	5.6234735	5.6883567
0.75	10^{-5}	5.6832576	5.6883567
0.90	10^{-3}	13.2777061	17.8517118
0.90	10^{-4}	17.0593987	17.8517118
0.90	10^{-5}	17.7604812	17.8517118

The $s = 0.6$ row at the smallest phase slightly overshoots its limit; this can reflect finite-phase corrections and grid error. Near $s = 1$ the fixed-exponent coefficient is large, and the critical crossover makes convergence slower. These observations are precisely why the numerical data are not used as a substitute for the integral and perturbation proofs.

9.4 Identity checks and finite-size behavior

At 70-digit working precision, the run checked equation (4.4) at 120 combinations of base, exponent, and position, including points close to the endpoints. The maximum absolute residual was below 1.3×10^{-67} . The independent cutoff integral for equation (4.19) differed from $2 \log(2/\pi)$ by about 2.31×10^{-15} , consistent with its endpoint cutoff of 10^{-15} . The beta/digamma representation of J_s was checked at $s = 0.6, 0.75, 0.9$; raw quadrature becomes less accurate near its integrability boundary, and the largest observed absolute discrepancy was about 3.18×10^{-7} . The exact symbolic identity checked was $\det(zI - J_b(u)) = z^2 + uLz - a_b d$.

For $b = 2$, $u = 0$, and $\tau = 1$, the finite-size limit in equation (6.7) is approximately 0.3504791. Using $c = 1/n^2$, the computed values of $n^{-1} 2^n M_{1,n}(1/n^2)$ at $n = 50, 100, 200, 400$ were approximately 0.3443066, 0.3421737, 0.3435866, and 0.3458026. The sequence need not approach its limit monotonically. The full output also contains detunings $u = -1, 1$, pressure crossover checks at $u = -2, -1, 0, 1, 2$, and mesh-doubling comparisons.

A representative mesh comparison at phase 10^{-5} changes the computed pressure by a few 10^{-7} . Such a change becomes a few hundredths after division by $|c|$. Therefore many displayed digits are useful for reproducing the computation but should not be interpreted as proven digits of the limiting continuum pressure.

9.5 Rebuilding the diagnostics

From the package root, a full run is

```
python -m pip install -r requirements.txt
OPENBLAS_NUM_THREADS=1 python code/verify.py --full
```

The second line is a POSIX shell example; on Windows set the environment variable using the shell's own syntax or omit it. The variable limits linear-algebra threading, not mathematical precision. Omitting `--full` runs a smaller diagnostic set. The script records library versions in its JSON output and regenerates the two numerical TeX tables.

10 A further-research agenda

Research question 1 (The remaining subcritical interval). Does equation (2.17) remain valid for $0 < s \leq 1/2$? The translation integral is already known there from lemma 7.2; the task is to estimate the spectral remainder using localization rather than its global operator norm. A useful first goal is an $o(|c|)$ bound for the second reduced-resolvent term. A counterexample would also be informative, since the present proof does not predict a transition at $s = 1/2$.

Research question 2 (The next critical term). Determine the next term after $\kappa_b \sqrt{|c|}$. Is there a $|c| \log |c|$ correction, an ordinary $|c|$ correction, or both? This requires resolving the other central matrix entries beyond the coarse Hölder translation bound and retaining the second-order chart correction. An explicit coefficient would explain the slow approach in the critical numerical table.

Research question 3 (A uniform matching region). Extend theorem 2.3 from bounded u to a quantitatively controlled region $|u| \leq U(t)$ with $U(t) \rightarrow \infty$. The factor $\exp(2ut \log t)$ in equation (6.2) indicates one of the natural restrictions. A theorem of this kind would turn equation (8.4) from a consistency check into a rigorous matched asymptotic expansion.

Research question 4 (Two-point endpoint regularity). Is $P_{b,1}$ actually $C^{0,1/2}$ on a whole neighborhood of zero? Corollary 2.2 only compares $P(c)$ with $P(0)$. Similarly, for fixed $1/2 < s < 1$, can one prove local Lipschitz regularity and smoothness away from the cusp, rather than only the one-sided first-order

expansion? Control of translated singularities at two nearby nonzero phases is needed to answer these stronger questions.

Research question 5 (Nonatomic phases). Classify phase regularity near nonzero rational phases and near irrational phases. The fixed-point evaluation that exposes equation (5.5) is special to the atomic phase. Periodic orbits of zeros may lead to different finite-dimensional obstructions, while recurrent nonperiodic zero orbits may require a different functional setting.

Research question 6 (Beyond consecutive digital masks). Find hypotheses on a trigonometric mask that imply an isolated Jordan chain at a critical moment exponent. If its nilpotent coupling is a and the leading lower-left perturbation is $d|c|^\rho$, the finite matrix predicts a leading $\sqrt{ad}|c|^{\rho/2}$. The research task is to derive those hypotheses and prove the required spectral separation for actual masks, rather than infer it from the matrix alone.

Research question 7 (Long rescaled times and higher-order semigroups). Obtain error bounds in equation (2.15) that remain useful for τ growing with $1/t$. Determine a next-order effective matrix and whether it produces a uniform correction to both the pressure and the finite-size moment. This would connect the compact-time sinh law to the single-eigenvalue regime quantitatively.

Research question 8 (Eigenfunctions, eigenmeasures, and equilibrium measures). The eigenprojection amplitudes diverge like $|c|^{-1/2}$, while their combination has a finite rescaled operator limit. Determine normalized weak limits and boundary layers of the individual left and right eigenvectors, and translate them into statements about equilibrium measures. The operator limit alone does not establish a variational classification of such measures.

Research question 9 (Certified spectral numerics). Replace the collocation diagnostics by explicit operator bounds and interval arithmetic. Useful deliverables would be rigorous enclosures for $P_{b,s}(c)$ at nonzero phases, certified constants in the error of equation (2.10), and mesh-independent upper and lower pressure bounds. Non-normality near the Jordan point makes residual-only certification especially inadequate.

Research question 10 (A staged Lean or Rocq formalization). Formalize first the digital-mask coboundary, the finite-dimensional Jordan algebra, the exact evaluation equation (5.5), and the matrix exponential calculation. Then formalize the weighted quadrature estimates, bounded projections, and the resolvent compression lemma. Only after these analytic bridges are present should the pressure theorem be declared formally verified. This route fits ProveIt's existing separation between executable checks and proved semantic connections.

11 Conclusion

The atomic phase is not merely a point where a fractional exponent formula needs a limiting interpretation. At the critical exponent the spectral geometry changes: a simple dominant branch and an atomic branch form a defective eigenvalue. The phase perturbation opens a coupling in the missing matrix direction, forcing a square-root splitting with the exact constant $\sqrt{2(b-1)\log b}$.

The argument supplies more than a scalar asymptotic. It gives an explicit Jordan chain, a directly proved complementary gap, a two-parameter collision law, and an operator finite-size crossover. Below the critical exponent, a different but compatible calculation yields an explicit linear cusp for $1/2 < s < 1$. The remaining interval is reduced to a specific localized perturbation problem rather than hidden inside an unsupported extrapolation.

A Proof-dependency and claim audit

The logical dependencies are as follows. The sampling inequality links integrals to cylinder pressure independently of any spectral claim. The special-function identity constructs the critical invariant plane.

Weighted quadrature and a fixed-iterate Hölder estimate prove its complementary gap; no general Perron theorem is used to assume that gap. A Riesz-projection chart then reduces the small-phase problem to a matrix with a controlled quadratic remainder. The exact fixed-point response determines the missing entry. Only after a nonzero positive moment amplitude is proved is the leading eigenvalue identified with pressure.

For the subcritical theorem, the order is separate: prove a simple atomic gap, identify its dual functional, evaluate a translated-zero integral, bound the operator remainder by $o(|c|)$, and then pass to pressure. The scalar first-order integral is not used as a substitute for the remainder estimate.

Potential failure mode	How it is excluded here
Confusing moment order and pressure order	$s = 2q$ is fixed in equation (2.8) ; the critical order is $q = 1/2$.
Using a normalized transfer operator	Equations (2.2) and (2.7) state the unnormalized operator and its exact integral factor.
Assuming a gap despite zeros of the weight	The gap is proved by weighted quadrature and a Hölder recurrence.
Losing the decisive perturbation coefficient	The compression lemma has an explicit norm-square error, and the lower-left entry is computed exactly.
Confusing spectral radius with pressure	A polynomial sampling bound and a nonzero positive moment amplitude supply the bridge.
Extrapolating the subcritical formula too far	The theorem is restricted to $s > 1/2$, precisely where the norm-square error is $o(c)$.
Treating numerics as a proof	All infinite statements have analytic proofs; collocation is labeled diagnostic throughout.
Claiming full Hölder regularity from a pointwise law	The regularity corollary explicitly states only pointwise optimality at zero.

B Reproduction and package contents

The package contains `article.tex`, its compiled `article.pdf`, the verification program, the actual numerical output, two generated numerical tables, a requirements file, a Makefile, and separate source and validation notes. No external paper or font file is redistributed. The document uses ordinary installed TeX packages, including Libertinus for typesetting.

To rebuild the article without rerunning the diagnostics, execute

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is embedded in the TeX source; no BibTeX download is needed. The source depends on the two included table files in `data/`. A full diagnostic rerun regenerates those tables before compilation. All new results in this package remain unrefereed and unformalized, regardless of the formal status of other ProveIt projects.

References

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